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AGRICULTURAL AND BIOSYSTEMS ENGINEER LICENSURE EXAMINATION
2024 PRE-BOARD EXAM (POST-TEST)

AREA III: *AB Structures, Environment Engineering, Bioprocess Engineering
and Allied Subjects*

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1. What is the power available from a water stream with an average width of 5 m, average depth 1.5 m, and with a flow velocity of 25 ft/min. The height of head is 1.64 yards from forebay down to turbine. Assume efficiency of 90.5%.

- a. 17.0 hp
- b. 18.8 hp
- c. 12.7 hp
- d. 14.0 hp

2. Compute the maximum total head at which 15.3-hp hydropower can be produced from the falling water originating from a square channel with sides of 1.64 ft and stream velocity of 0.36 kph. Assume eff. of 57.5%.

- a. 8 m
- b. 9 m
- c. 5 m
- d. 6 m

3. What change in volume results if 60.0 mL of gas is cooled from 33.0 °C to 5.00 °C?

- a. 54.51 mL
- b. 5.49 mL
- c. 53.2 mL
- d. 6.8 mL

4. The maximum thermal efficiency possible for a power cycle operating between 1200°F and 225°F is:

- a. 58.54%
- b. 57.81%
- c. 56.01%
- d. 58.73%

5. The walls of a cold storage plant are composed of an insulating material ($k=0.2336$ kJ/hr-m-°C) 10.16 cm thick held between two layers of concrete ($k=3.7382$ kJ/hr-m-°C) each 10.16 cm thick. The film coefficients are 81.76 kJ/hr-m²-°C on the outside and 40.88 kJ/hr-m²-°C on the inside. Cold storage temperature is -6.7°C and the ambient temperature is 32°C. Determine the heat transmitted in kW through an area of 55.74 m².

- a. 1.14 kW
- b. 1.74 kW
- c. 1.81 kW
- d. 1.91 kW

6. At an average temperature of 100°C, hot air flows through a 2.5 m long tube with an inside diameter of 50 mm. The temperature of the tube is 20°C along its length. Convective heat transfer coefficient is 20.1 W/m²-K. Determine the heat transfer from air to the tube.

- a. 624 W
- b. 909 W
- c. 900 W
- d. 632 W

7. Water at a mean temperature of 150°F flows through a steel pipe having an inside diameter of 1 and an outside diameter of 1-1/4". The outer surface of the pipe is exposed to air at 70°F. The heat transfer coefficients on the water and air sides are 100 Btu/ft²-hr-°F and 1 Btu/ft²-hr-°F, respectively and k for steel may be taken as 25 Btu/ft-hr-°F. The heat transfer rate per foot of pipe is:

- a. 13.2 Btu/ft-hr
- b. 19.2 Btu/ft-hr
- c. 25.9 Btu/ft-hr
- d. 69.2 Btu/ft-hr

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8. Compute for the heat transfer in W for a triangular surface (emissivity of ϵ) with $3/2$ m by ϵ m for base and height, respectively. Change in temperature is from 20°C to 100°C .

- a. 12743496.35 W
- b. 1274349.635 W
- c. 127434.9635 W
- d. 127434963.5 W

9. The rate of convection of a boiling water is E watt. Then the rate of convection of the water at 50°C will be:

- a. $1/2 E$
- b. $1/4 E$
- c. $2/3 E$
- d. $1/3 E$

10. To what pressure must a gas be compressed in order to get into a 3.00 cubic foot tank the entire weight of a gas that occupies 400.0 cu. ft. at standard pressure?

- a. 133 atm
- b. 135 atm
- c. 143 atm
- d. 123 atm

11. The primary law governing blackbody radiation that quantifies the intensity of radiation emitted by unit surface area into a fixed direction from the blackbody as a function of wavelength for a fixed temperature.

- a. Stefan-Boltzmann Law
- b. Newton's Law of Cooling
- c. Fourier's Law
- d. Planck's Law

12. In CPES rating, what is the percentage for Materials upon completion?

- a. 0.20
- b. 0.15
- c. 0.40
- d. 0.30

13. What is the size of the wire AWG 10?

- a. 5.0 mm^2
- b. 5.5 mm^2
- c. 6.0 mm^2
- d. 8.8 mm^2

14. Amount of heat added or removed measured in change of temperature without changing state.

- a. Latent Heat
- b. Sensible Heat
- c. Specific Heat
- d. Innate Heat

15. Product of the coefficient of hulling and the coefficient of wholeness of grains, expressed in percent.

- a. Coefficient of wholeness
- b. Coefficient of hulling
- c. Hulling Efficiency
- d. Fineness Modulus

16. The amount paid to project owners from Clean Development Mechanism (CDM) fund for carbon emission reduction projects.

- a. Carbon credit value
- b. Emission value
- c. Carbon trading value
- d. Carbon emission value

17. What is the peak amperage of a solar battery charging station equipped with 64 units of 80 watt-peak 5.25-ampere solar module? The system is divided into 8 charging channels, each channel has 8 modules connected in parallel.

- a. 24.0 A
- b. 42.0 A
- c. 16.6 A
- d. 21.0 A

18. What is the minimum methane content of biogas in order to operate biogas-fueled generators?

- a. 50%
- b. 65%
- c. 85%
- d. 90%

19. Type of biogas plant which have aerobic conditions in the upper layers and anaerobic processes occurring in the bottom layers, especially in the settled solids?

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- a. Aerobic Biogas
- b. Anaerobic Biogas

- c. Facultative Biogas
- d. Activated Biogas

20. It is a wastewater, either treated or untreated, discharged from a reservoir, basin, or treatment plant.

- a. Effluent
- b. Influent

- c. Turbid Water
- d. Hard Water

21. In PAES 421:2009, the tank shall have heating device able to heat a complete charge of bituminous liquid to ____ °C.

- a. 80
- b. 120

- c. 180
- d. 200

22. In PAES 415:2001, under natural ventilating systems of greenhouse, large vent openings provide the most ventilation with total vent area that should be ____ - ____ of floor area.

- a. 15% - 30%
- b. 30% - 50%

- c. 25% - 50%
- d. 15% - 25%

23. In PAES 421:2009, it is constructed to drain water from the subgrade and to collect surface water either from the roadway surface or adjacent.

- a. roadside ditch
- b. culvert

- c. channel
- d. conduit

24. In PAES 408:2001, the minimum space requirement for carabao feedlot feeding shall be ____ m/animal.

- a. 0.5
- b. 1.0

- c. 0.75
- d. 1.25

25. The maximum number of outlets for a 15-A circuit.

- a. 8-10 outlets
- b. 10-12 outlets

- c. 14-15 outlets
- d. 6-8 outlets

26. Dissolved oxygen concentrations will be the lowest within a waterbody on ____.

- a. Afternoon
- b. Sunrise

- c. Sunset
- d. Noon

27. In PAES 401:2000, heaters shall be installed in creep areas to provide newborn pigs its required temperature of ____°C - ____°C until they are 3 days old.

- a. 32-35
- b. 29-35

- c. 30-35
- d. 27-35

28. What is the mechanical power of a windmill with a diameter of 15 meters and velocity of 7 mps?

- a. 6.160 kW
- b. 6.601 kW

- c. 6.610 kW
- d. 6.061 kW

29. A Carnot engine is working between two reservoirs with temperatures of 1000°C and 500°C, what is the thermal efficiency of the system?

- a. 40%

- c. 50%

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- b. 30% d. 100%
30. It is an environmental-friendly refrigerant.
- a. CFC c. HCFC
b. HFC d. FC
31. It is the programming language of Arduino microcontrollers.
- a. R c. C/C++
b. Python d. Java
32. In Arduino programming, the meaning of IDE is:
- a. Integrated Development Environment c. International Development Environment
b. Internal Development Environment d. Interconnected Development Environment
33. The Feed-in-Tariff (FIT) of wind energy in the BiGSHOW is:
- a. 6.63 PHP/kW-hr c. 5.90 PHP/kW-hr
b. 9.68 PHP/kW-hr d. 8.53 PHP/kW-hr
34. It is the gas produced from agricultural waste which is odorless, tasteless, colorless and flammable.
- a. CH₄ c. O₂
b. CO d. CO₂
35. All are products of pyrolysis except:
- a. Vinegar c. Oil
b. Tar d. Syngas
36. What is the electrical energy consumption of a 20-hp vermicast production shredder plugged in a 220-V source if it requires 3 hrs to shred 1.8 tonnes of farm waste?
- a. 0.205 kW-h/kg c. 0.250 kW-h/kg
b. 0.025 kW-h/kg d. 0.052 kW-h/kg
37. It is a type of whitening machine consisting of a ribbed cylinder enclosed in a perforated steel housing.
- a. friction type c. rubber roll type
b. abrasive type d. cone "cono" type
38. What is the input current to a 6-hp single phase electric motor use to drive a hammer mill at 2500 rpm. The input voltage is 230 volt. Assume a phase angle of 30 degrees.
- a. 20.34 amp c. 27.24 amp
b. 19.46 amp d. 22.47 amp
39. When a system deviates infinitesimally from equilibrium at every instant of its state, it is undergoing:
- a. isobaric process c. isometric process
b. quasi-steady process d. isentropic process
40. A throw-in type of thresher wherein cut plants are fed between the rotating cylinder and stationary concave, and the threshed materials/straws are discharged out of the threshing chamber tangentially.
- a. radial type thresher c. axial-flow thresher

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b. hold-on type thresher

d. through flow thresher

41. A wooden pole (SG=0.7), 13 cm by 13 cm by 8 m long, hangs vertically from a string in such a way that 5 m is submerged in water and 3 m is above the surface. What is the tension in the string?

a. 10.14 N

c. 50.7 N

b. 99.47 N

d. 431.05 N

42. How much energy is needed for a 12-volt source to move 5 coulombs of charge?

a. 20 joules

c. 0.6 joules

b. 2.4 joules

d. 60 joules

43. It is a hypothetical activity which takes no resource or time to complete. It is represented by broken arrowed line & is used for either distinguishing activities having common starting & finishing events or to identify & maintain proper precedence relationship between activities that are not connected by events

a. Free Float

c. Dummy activity

b. Slack Activity

d. Zero Slack

44. In lateral and sub-lateral canal where the available roadway width is 3 meters, the carriageway width shall be 2 meters and the shoulder width shall be 0.5 meters. If the available roadway width is 4 meters, the shoulder shall expand to 1 meter.

a. roadway on main irrigation canals

c. roadway on lateral & sub lateral irrigation canals

b. road bed lateral canals

d. turn-out section canals

45. When you want to use objects from a drawing file in another application, you can cut these objects to the Clipboard and then paste them into another application. CUTCLIP copies selected objects to the Clipboard, removing them from the drawing using the shortcut:

a. Ctrl Y

c. Ctrl X

b. Ctrl Z

d. Ctrl V

46. In AutoCAD, it creates copies of objects arranged in a pattern. You can create copies of objects in a regularly spaced rectangular, polar, or path.

a. Ortho

c. Array

b. Offset

d. Fillet

47. A structure the consists of compressive force action and supports only axial loadings is:

a. Beam

c. Purlins

b. Column

d. Girder

48. A beam needs 4 pcs of 25 mm RSB. If there is no available 25 mm RSB, how many 22 mm RSB can substitute for the 25 mm?

a. 4 pcs

c. 6 pcs

b. 5 pcs

d. 7 pcs

49. How many pieces of umbrella nails are there if there are 20 kilograms of umbrella nails?

a. 20,000 pcs

c. 24,000 pcs

b. 21,000 pcs

d. 12,000 pcs

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50. It a positively charged particle made up of two protons and two neutrons and released by a radioactive nucleus.

- a.** Alpha radiation
- b.** Beta radiation
- c.** Ultra-violet radiation
- d.** Gamma radiation

51. Given the initial temperature of 50 F and initial pressure of 20 bar, the pressure changed to 58 bar, determine the change in temperature.

- a.** 696
- b.** 969
- c.** 1910
- d.** 1019

52. It is the production of food, fiber, or other plant or animal products using farming techniques that protect the environment, public health, human communities, and animal welfare.

- a.** Organic Agriculture
- b.** Sustainable Agriculture
- c.** Traditional Agriculture
- d.** Conventional Agriculture

53. It is a production system that sustains the health of soils, ecosystems and people. It relies on ecological processes, biodiversity and cycles adapted to local conditions, rather than the use of inputs with adverse effects.

- a.** Organic Agriculture
- b.** Sustainable Agriculture
- c.** Traditional Agriculture
- d.** Conventional Agriculture

54. It is the layer of aggregate, soil-treated aggregate, treated soil, or soil aggregate that rests upon the subbase or if no subbase, upon the subgrade. Treatment may include application of chemical-based soil additives such as soil-stabilizers and/or any approved method.

- a.** Aggregate Base Course
- b.** Base Course
- c.** Subbase Course
- d.** Aggregate Subbase Course

55. It involves the arrangement of compost mix in long, narrow piles or windrows that are periodically turned to maintain aerobic conditions.

- a.** Aerated static pile
- b.** In-vessel system composting
- c.** Windrow composting
- d.** Bin composting

56. The percent of moisture content of test materials for peanut sheller.

- a.** 8%-10%, wet basis
- b.** 12%-14%, wet basis
- c.** 8%-10%, dry basis
- d.** 12%-14%, dry basis

57. Measures velocity by the change of source light or sound frequency from the frequency of reflections from moving particles.

- a.** Optical Strobe
- b.** Doppler
- c.** Bellow
- d.** Platometer

58. Ratio of the weight of corn kernels that fell out from the machine during shelling operation to the weight of the total corn kernel input of the sheller, expressed in percent.

- a.** Scattering loss
- b.** Cylinder loss
- c.** Separation loss
- d.** Blower loss

59. A 3.5-tonne capacity *Jatropha* oil expeller was loaded full with dried *Jatropha* seeds in 5 minutes. It produced 800 kg of crude oil in 4 hours. What is its crude oil production rate?

- a.** 300 kg/h
- b.** 200 kg/h
- c.** 600 kg/h
- d.** 400 kg/h

60. A 25MW-100MW hydro power plant is classified as: _____.

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- a. Micro hydro
 - b. Small hydro
 - c. Mini hydro
 - d. Medium hydro

61. Convert the hexadecimal number DECAF to octal number system.

 - a. 3636257₈
 - b. 3366257₈
 - c. 3663247₈
 - d. 3362347₈

62. Which one is an integrated circuit used in instrumentation and control?

 - a. TD 8062
 - b. LM333
 - c. ES8813
 - d. CQ 556

63. In PAES 421:2009, the minimum radius of curvature for flat, rolling and mountainous terrains respectively shall be.

 - a. 75, 50 & 5 m
 - b. 500, 150 & 50 m
 - c. 150, 100 & 50 m
 - d. 600, 200 & 100 m

64. It refers to the extent or degree by which the bran layer of the brown rice is removed as a result of whitening.

 - a. Whitening degree
 - b. Milling degree
 - c. Milling recovery
 - d. Milling index

65. The most efficient and ideal gas cycle.

 - a. Rankine Cycle
 - b. Carnot Cycle
 - c. Otto Cycle
 - d. Brayton Cycle

66. Standard thickness of concrete of trapezoidal ditch w/ riprap.

 - a. 0.25 m
 - b. 0.15 m
 - c. 0.30 m
 - d. 0.20 m

67. Compute for the mounting height in a warehouse with 5x5 grid lighting layout with mounting height ratio of 1.0

 - a. 4 m
 - b. 25 m
 - c. 5 m
 - d. 8 m

68. It is a rice grain from which the hull, the germ, the outer bran layers, and the greater part of the inner bran layer have been removed, but part of the lengthwise streaks of the bran layers may still be present on less than 20% of the sample grains.

 - a. Well-milled rice
 - b. Regular milled rice
 - c. Overmilled rice
 - d. Undermilled rice

69. Moisture gradient of corn and rice in mechanical dryer.

 - a. 0.5%
 - b. 1.0%
 - c. 2.5%
 - d. 2.0%

70. It is the chamber maintained under pressure for uniform distribution of the heated air through the grain mass.

 - a. dust collection system
 - b. drying chamber
 - c. plenum
 - d. tempering bin

71. The rate of radiation of a black body at 0°F is 2E watt. Then the rate of radiation of this black body at 460°F will be:

 - a. 32E
 - b. 16E
 - c. 8E
 - d. 4E

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72. In PAES 420:2002, the door height shall be provided with a clearance of at least ____ m above the tallest machinery.

- a.** 0.3
- b.** 0.4
- c.** 0.6
- d.** 0.7

73. Heat absorbed by a unit mass of a material at its boiling point to convert the material into a gas without temperature change.

- a.** Latent heat of sublimation
- b.** Latent heat of deposition
- c.** Latent heat of fusion
- d.** Latent heat of vaporization

74. Which of the following is a good insulator?

- a.** Brass
- b.** Mica
- c.** Tin
- d.** Platinum

75. The angle measured from the horizontal at which product will start to move downwards over a smooth surface with gravity discharging the product is called a _____.

- a.** angle of discharge
- b.** angle of friction
- c.** angle of repose
- d.** hopper angle

76. In PAES 616:2016, precautionary measures in *Wastewater Re-Use for Irrigation*, which is false?

- a.** Direct contact with wastewater shall be avoided
- b.** Use of fine mist for sprinkler irrigation shall be used
- c.** Potable and wastewater lines shall not cross connect
- d.** Irrigation with wastewater shall be stopped immediately when algal bloom occurs

77. It is the milled corn kernels with particle size between 1.2 mm to 1.4 mm.

- a.** Grit #16
- b.** Grit #18
- c.** Grit #10
- d.** Grit #14

78. The amount of moisture in the grain expressed as a percentage of the total weight of the samples.

- a.** Moisture content dry basis
- b.** Moisture content wet basis
- c.** Moisture content
- d.** Equilibrium moisture content

79. Recommended stacking density for corn in terms of number of bags per cubic meter.

- a.** 10
- b.** 9
- c.** 12
- d.** 15

80. A 1m x 10 m poultry handling area is to be lighted at an average maintained footcandle of 50 fc. How many 2-lamp fixtures of 120-cm long F40 T12 WW rapid start fluorescent lamps are required assuming the CU is 0.35 and the LLF is 0.6?

- a.** 3 fixtures
- b.** 6 fixtures
- c.** 2 fixtures
- d.** 4 fixtures

81. The heat exchange when a material changes its state of molecular aggregation without change in temperature.

- a.** Latent heat
- b.** Specific heat
- c.** Sensible heat
- d.** All of the above

82. A systematic approach to identifying, evaluating, and controlling food safety hazards.

- a. CPES
b. HACCP

- c. GMP
d. SSOP

83. For a purely capacitive circuit, the power factor is:

- a. 0 (lagging)
b. <1

- c. >1
d. 0 (leading)

84. What force is required to punch a 30 mm diameter hole in a plate that is 15 mm thick? The shear strength is 424.4 MPa.

- a. 550 kN
b. 600 kN

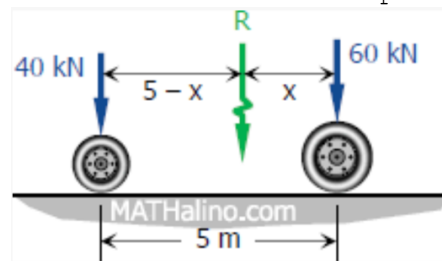
- c. 500 kN
d. 650 kN

85. In PAES 418:2002, area in sqm per ton of commodity in primary processing plant is:

- a. 10 sqm per ton
b. 15 sqm per ton

- c. 20 sqm per ton
d. 25 sqm per ton

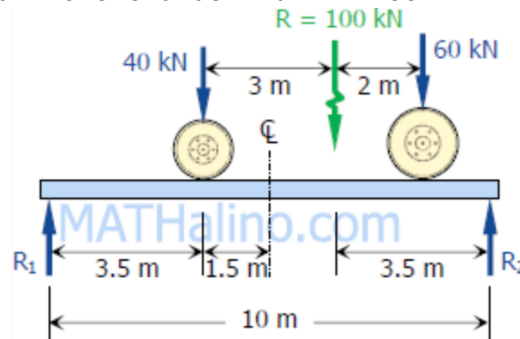
86. Compute for the value of x for a truck with axle loads of 40 kN and 60 kN on a wheel base of 5 m rolls across a 10-m span.



- a. 3
b. 2

- c. 2.5
d. 1.5

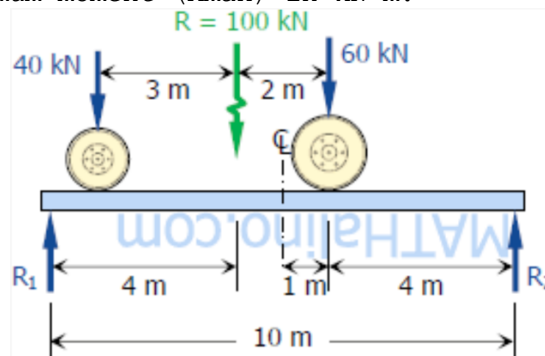
87. Compute for maximum moment under 40 kN wheel in kN-m.



- a. 160
b. 245

- c. 122.5
d. 320

88. Compute for maximum moment (M_{max}) in kN-m.



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- a. 160
- b. 245
- c. 122.5
- d. 320

89. In a capacitive circuit, the capacitance is 0.5 μF . What is the capacitive resistance at 60-Hz source?

- a. 188.5 $\text{k}\Omega$
- b. 5.305 $\text{k}\Omega$
- c. 1.885 $\text{k}\Omega$
- d. 53.05 $\text{k}\Omega$

90. Corn initially at ____ % is to be dried to 14%. If the initial weight of the shelled corn is 1000 kg, its final weight is 884 kg. Compute for the initial MC.

- a. 28.93%
- b. 24.89%
- c. 23.98%
- d. 29.38%

91. In PAES 411:2000, the gut room/tripery shall be provided with lighting intensity of _____ lux.

- a. 200
- b. 220
- c. 540
- d. 110

92. Convert 25% MCdb to MCwb.

- a. 30.00%
- b. 33.33%
- c. 20.00%
- d. 16.67%

93. In PAES 409:2002, how many milking operators are required for a milking herd having 21-30 animals without a milking machine?

- a. 1
- b. 2
- c. 3
- d. 4

94. Determine the volume of the digester required to produce biogas for a livestock farm having 500 heads of cattle. The projected retention time is 25 days.

- a. 175 m^3
- b. 350 m^3
- c. 424 m^3
- d. 438 m^3

95. What is the equivalent size in mm^2 of a 300 MCM cable?

- a. 152 mm^2
- b. 155 mm^2
- c. 125 mm^2
- d. 132 mm^2

96. Burning efficiency of the biomass furnace shall be at least:

- a. 98%
- b. 95%
- c. 90%
- d. 80%

97. Stoichiometric air requirement for pyrolysis.

- a. 10-20%
- b. 20-30%
- c. 30-40%
- d. 40-50%

98. The C/N Ratio of agricultural waste.

- a. 10:20
- b. 20:30
- c. 30:40
- d. 40:50

99. Main product of gasification.

- a. CH_4
- b. CO
- c. O_2
- d. CO_2

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100. The heating system efficiency for rice hull direct-fired mechanical grain dryer.

- a. 95%
- b. 80%
- c. 90%
- d. 65%

101. What is the difference of boiling water at higher elevation compared to lower elevation?

- a. At higher elevation, pressure is lower and ambient temperature is higher : slower heating rate leading to longer boiling time
- b. At higher elevation, pressure is higher and ambient temperature is lower : faster heating rate leading to longer boiling time
- c. At higher elevation, pressure is higher and ambient temperature is lower : faster heating rate leading to faster boiling time
- d. At higher elevation, pressure is lower and ambient temperature is lower : slower heating rate leading to longer boiling time

102. An automobile tire is inflated to 32 psig pressure at 50°F. After being driven, the temperature rises to 75°F. Assuming that the volume remains constant, the difference in gage pressure is:

- a. 1.6 psig
- b. 23.4 psig
- c. 2.3 psig
- d. 16.0 psig

103. It is a cantilever beam that holds the gasholder/movable cover in position at the desired biogas pressure

- a. gasholder retainer
- b. gasholding beam
- c. gasholder frame
- d. biogas cover-lever

104. Biochemical Oxygen Demand (BOD₅) in mg/L for wastewater re-use:

- a. <500
- b. <200
- c. <150
- d. <50

105. What is the temperature of 2 liters of water at 30°C after 0.1 Mcal of heat have been added to it?

- a. -20.0°C
- b. 80.0°C
- c. 25.0°C
- d. 35.0°C

106. The height of hover type brooder shall be adjustable. Hovers shall be maintained at a minimum clearance of ____ above the back of the birds.

- a. 120 mm
- b. 150 mm
- c. 170 mm
- d. 200 mm

107. Doe/Ewe of 70 kg space requirement in Slotted Flooring.

- a. 0.7 sq. m.
- b. 0.9 sq. m.
- c. 1.1 sq. m.
- d. 2.5 sq. m.

108. In housing for broiler, the width of pen shall be about ____ m.

- a. 10-12
- b. 12-14
- c. 14-16
- d. 16-18

109. The fence height for Brahman and other exotic breeds in a cattle ranch shall be ____.

- a. 1.7 - 1.8 m
- b. 1.8 - 2.0 m
- c. 1.5 - 1.7 m
- d. 2.0 - 2.2 m

110. The lairage for small and large animals shall be located at least ____ away from the slaughterhouse.

- a. 20 m
- b. 50 m
- c. 10 m
- d. 30 m

111. Maximum allowable noise limits of a slaughterhouse during night located in a noise sensitive place is ____.

- a. 55 dB
- b. 50 dB
- c. 40 dB
- d. 45 dB

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135. Defects Liability Period shall be _____ year/s from project completion up to final acceptance by the Government.

- a. 1 year
- b. 2 years
- c. 3 years
- d. 5 years

136. What is a plan of action proposed by the constructor to the concerned agency's implementing office which is aimed to prevent the recurrence of non-conformance findings on his on-going and/or future projects.

- a. Preventive Correction Plan
- b. Corrective Action Plan
- c. Sustainability Action Plan
- d. Preventive Action Plan

137. A step at which control can be applied and is essential to prevent or eliminate a food safety hazard or reduce it to an acceptable level.

- a. Critical Limit
- b. Critical Control Point
- c. HACCP Systems
- d. Hazard Analysis

138. Any step at which biological, chemical, or physical factors can be controlled.

- a. Hazard Point
- b. Critical Point
- c. Control Point
- d. Corrective Point

139. Also called exhaust gas:

- a. Flue gas
- b. Syngas
- c. Biogas
- d. Producers gas

140. Most common method for converting biomass to useful energy

- a. Gasification
- b. Pyrolysis
- c. Combustion
- d. Liquefaction

141. Crude oil is mainly composed of:

- a. Carbon
- b. Hydrocarbon
- c. Nitrogen
- d. C/N

142. A rigid tank contains 5 kg of an ideal gas at 4 atm and 40°C. Now a valve is opened, and half of mass of the gas is allowed to escape. If the final pressure in the tank is 1.5 atm, the final temperature in the tank is:

- a. -38°C
- b. -40°C
- c. 38°C
- d. 40°C

143. Solar Powered Irrigation System Photovoltaic array should be facing _____ based on efficiency and performance

- a. North
- b. South
- c. East
- d. West

144. Scopes of CPES includes the following except:

- a. public infrastructures
- b. private infrastructures
- c. local constructors
- d. foreign constructors

145. Converter of energy into other form

- a. Transformer
- b. Transducer
- c. Transmutator
- d. All of the Above

146. A rice miller buys 10 kg of dried and clean palay at P13/kg. Assuming that his cost of operation is P4/kg milled rice, how much should he sell (P/kg) to break even. Neglect by-products and assume milling recovery of 64%.

- a. 20.4
- c. 22.4

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- b. 24.3 d. 23.3
147. The energy received by or projected onto a surface, expressed in $W \cdot m^{-2}$
- a. albedo c. radiosity
b. radiant energy d. insolation
148. What is the density (kg/m^3) of dry air at 101 kPa and $2^\circ C$? Gas constant = 287 J/kg-K for air and 462 J/kg-K for water.
- a. 1.28 c. 1.08
b. 1.18 d. 1.38
149. What is the value of the work done for a closed, reversible, isometric system?
- a. positive c. zero
b. negative d. cannot be determined
150. Consider 100 m^3 of moist air at 101.325 kPa, $35^\circ C$, 70% relative humidity. If saturation pressure of water is 5.628 kPa at $35^\circ C$, then partial pressure of dry air is:
- a. 5.69 c. 3.94
b. 9.74 d. 8.73
151. In psychrometric chart, these represented by the x-axis.
- a. Dry Bulb Temperature c. Specific Enthalpy
b. Wet Bulb Temperature d. Specific Volume
152. This law states that if two thermodynamic systems are in thermal equilibrium with a third, they are also in thermal equilibrium with each other.
- a. 1st Law of Thermodynamics c. 3rd Law of Thermodynamics
b. 2nd Law of Thermodynamics d. 0th Law of Thermodynamics
153. In PAES 421:2009, what is the minimum carriage way of one-lane and two-lane FMR, excluding shoulders?
- a. 2 and 4 m c. 2.5 and 5 m
b. 4 and 6 m d. 3 and 6 m
154. In PAES 402:2000, the recommended brooding temperature for 14-21 days chicks.
- a. provide heat when necessary c. $32-35^\circ C$
b. $29-32^\circ C$ d. $27-29^\circ C$
155. What is the defect of wood that causes abnormal growth which occurs at the starting point of a limb or branch of a tree?
- a. shake c. check
b. knot d. wane
156. In PAES 410:2000, the lairage shall be provided with lighting intensity of _____ lux.
- a. 200 c. 500
b. 100 d. 110
157. What diode is needed as protection against lightning in solar installations?
- a. Bypass diode c. Shading diode
b. Blocking diode d. Lightning diode
158. Compute for the heat transfer in BTU/hr for a rectangular surface (emissivity of $\cdot\cdot$) with $3/2$ ft by $\cdot\cdot$ ft for base and height, respectively. Change in temperature is from $20^\circ F$ to $100^\circ F$.
- a. 29.53 GBTU/hr c. 295.3 GBTU/hr
b. 0.2953 GBTU/hr d. 2.953 GBTU/hr

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159. In CPES rating, the maximum weight rating for Facilities criteria during construction.

- a. 0.20
- b. 0.15
- c. 0.40
- d. 0.03

160. In PAES 403:2001, under the individual nest subsection, there shall be one nest for each ____ to ____ layer.

- a. 5 - 6
- b. 8 - 10
- c. 7 - 8
- d. 4 - 6

161. A complete lighting unit consisting of a lamp or lamps together with parts designed to distribute the light, to position and protect the lamps, and to connect the lamps to the power supply.

- a. lumen
- b. luminaire
- c. luminous
- d. lighting fixture

162. What is the meaning of HAZOP?

- a. Hazard & Operations
- b. Hazard & Operability
- c. Hazard & Operatability
- d. Hazard & Operationalization

163. In CPES rating, the Satisfactory Qualitative Description have a corresponding CPES rating range of:

- a. 89% - 96%
- b. 75% - 82%
- c. 82% - 89%
- d. >96%

164. One (1) ton of refrigeration is equal to:

- a. 3.52 kW
- b. 1,000 kg
- c. 2,200 lbs
- d. 1 kW

165. In the development of HACCP plan, what should come first?

- a. Describe the food and its distribution
- b. Assemble a HACCP Team
- c. Describe the intended use and consumers of the food
- d. Verify the flow diagram

166. What cement class is used for the construction of FMR?

- a. AA
- b. A
- c. B
- d. C

167. What is the second principle of HACCP?

- a. Conduct a hazard analysis
- b. Establish monitoring procedures
- c. Establish critical limits
- d. Determine the CCP

168. According to HACCP, this is an example of a physical hazard.

- a. A piece of a glass in pizza
- b. Bone in a chicken
- c. Seed in a grape
- d. All of the above

169. What TESDA National Certificate Level is required for Agricultural and Biosystems Engineering Technicians and Operators for the postharvest processing of rice and other grain products?

- a. Drying and Milling Plant Servicing NC I
- b. Drying and Milling Plant Servicing NC II
- c. Drying and Milling Plant Servicing NC III
- d. Drying and Milling Plant Servicing NC IV

170. It refers to local activity or series of activities to maintain or raise the quality or change the form or characteristics of agricultural, fishery, forestry and biological products/materials and includes, but is not limited to, cleaning, sorting, grading, treating, drying, dehydrating, grinding, mixing, milling, canning, dressing, slaughtering, freezing, pasteurizing, conditioning, packaging, repacking, transporting of agricultural, fishery, forestry and other biological products/materials.

- a. Agricultural Process Engineering
- b. Agricultural and Bio-Processing
- c. Agricultural and Bioprocess Engineering
- d. Crop Process Engineering

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171. According to the National Building Code, the agricultural buildings are classified under Group ____.

- a. B
b. D
c. J
d. G

172. A gas occupies 12.3 liters at a pressure of 40.0 mmHg. What is the volume when the pressure is increased to 60.0 mmHg?

- a. 8.5 L
b. 8.2 L
c. 7.3 L
d. 9.0 L

173. The dimensionless number representing the ratio of momentum diffusivity over the thermal diffusivity

- a. Nusselt Number
b. Prandtl Number
c. Grashof Number
d. Reynolds Number

174. Calculate the amount of latent heat required to convert 3 kg of ice into water at 0 degrees Celsius.

- a. 334 kJ
b. 668 kJ
c. 1002 kJ
d. 1,336 kJ

175. How much heat in kJ, must be transferred to 20 kg of air to increase the temperature from 20°C to 280°C if the pressure is maintained constant:

- a. 5,200 kJ
b. 2,500 kJ
c. 2,550 kJ
d. 5,250 kJ

176. A sample of gas at 3.00×10^3 mmHg inside a steel tank is cooled from 500.0 °C to 0.00 °C. What is the final pressure of the gas in the steel tank?

- a. 1051 kPa
b. 150.10 kPa
c. 141.27 kPa
d. 1412 kPa

177. The atmospheric pressures at the top and bottom of a building are read by a barometer to be 96.0 and 98.0 kPa. If the density of air is 1.2 kg/m³, the height of the building is:

- a. 142 m
b. 210 m
c. 170 m
d. 153 m

178. In a refrigeration system, the equipment that rejects heat is the:

- a. Evaporator
b. Condenser
c. Compressor
d. Valve

179. In minimum, all culverts and storm drains used in relation with FMR should be designed using a storm of ____ and 24 hours.

- a. 1-year
b. 2-year
c. 3-year
d. 4-year

180. A 5-m³ rigid vessel contains steam at 20 MPa and 400°C. The mass of the steam is:

- a. 332 kg
b. 322 kg
c. 32 kg
d. 32.2 kg

Compute for the water power of the two renewable projects: waterwheel and waterfall sources summarized below:

Parameters	Waterwheel Specification	Waterfall Specification
Depth of Channel, ft	7	10
Width of Channel, ft	3	5
Water Flow Velocity, fps	5	7
Head, ft	20	30
Hydraulic Efficiency, %	60	45

181. Compute for the theoretical HP of WATERWHEEL Power Harnessing System

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- a. 230.53 HP c. 143.01 HP
b. 238.35 HP d. 141.13 HP
182. Compute for the theoretical HP of WATERFALL Power Harnessing System
a. 536.26 HP c. 1191.7 HP
b. 1911.7 HP d. 563.62 HP
183. Compute for the actual HP of WATERWHEEL Power Harnessing System
a. 230.53 HP c. 143.01 HP
b. 238.35 HP d. 141.13 HP
184. Compute for the actual HP of WATERFALL Power Harnessing System
a. 536.26 HP c. 1191.7 HP
b. 1911.7 HP d. 563.62 HP
185. In perspective projection, the horizontal plane in which the object is assumed to be situated is called _____
a. horizontal plane c. ground plane
b. picture plane d. auxiliary ground plane
186. To understand some of the hidden geometry of components an imaginary plane is used to cut the object which is called _____
a. auxiliary plane c. picture plane
b. section plane d. additional plane
187. If an isometric drawing is made use of isometric scale then the drawings are called _____
a. Isometric projection c. Isometric perception
b. Isometric view d. Orthographic view
188. A sphere is cut by a plane which is inclined to both reference planes the top view and front view of section will be _____
a. line, line c. ellipse, circle
b. circle, circle d. ellipse, ellipse
189. A grain or a piece of a grain with its length equal to or greater than three-fourths ($3/4$) of the average length of the whole kernels
a. paddy c. well-milled rice
b. head rice d. cargo rice
190. A type of a huller with rotating blades and utilizes pressure such as Coriolis' force, frictional force from the blades, or impact force at collision with the blades and the peripheral surface.
a. centrifugal type c. abrasive type
b. friction type d. cono type
191. A room is filled with saturated moist air at 30°C and a total pressure of 102 kPa and actual partial pressure of 3.3 kPa. If the mass of dry air in the room is 109 kg, the mass of water vapor is:
a. 2.36 c. 2.26
b. 2.64 d. 2.76
192. The refrigerating capacity of a R-12 system is 22 kW. Compressor power is 7.8 hp. Determine the COP of the refrigeration system.
a. 2.68 c. 3.94
b. 3.78 d. 2.51
193. A typical new household refrigerator consumes about 700 kWh of electricity per year and has a coefficient of performance of 1.4. The amount of heat removed by this refrigerator from the refrigerated space per year is:
a. 3,528 MJ/yr c. 9,320 MJ/yr
b. 2,024 MJ/yr d. 1,092 MJ/yr

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194. A room contains 50 kg of dry air and 0.6 kg of water vapor. The total temperature and pressure in the room are 32°C and 95 kPa. Given that the saturation pressure for water at 32°C is 4.19 kPa, the relative humidity of air in the room is:

- a. 49.3%
- b. 56.7%
- c. 42.9%
- d. 27.9%

195. A refrigerator is removing heat from a cold medium at 3°C at a rate of 7200 kJ/h and rejecting waste heat to a medium at 30°C. If the coefficient of performance of the refrigerator is 2, the power consumed by refrigerator is:

- a. 0.5 kW
- b. 0.1 kW
- c. 0.4 kW
- d. 0.3 kW

196. A room contains air at 30°C and a partial pressure of 2.9 kPa with a relative humidity of 87 percent. Given that the humidity ratio is 0.01428, the total pressure is:

- a. 85.80 kPa
- b. 90.6 kPa
- c. 129.26 kPa
- d. 132.5 kPa

197. Sherwin and Alex decided to live together. As part of their necessities, they decided to buy a new air conditioner. Their chosen air conditioner has an energy efficiency ratio (EER) of 12 kJ/W-h. What is its equivalent COP?

- a. 4.6
- b. 4.2
- c. 4.3
- d. 4.5

198. How many kilograms of rice hulls are needed to dry 1 ton of palay from 24% to 14% w.b.? Assume latent heat of vaporization is 2,500 kJ/kg, heating value of rice hull is 14 MJ/kg and efficiency of 100%.

- a. 20.8 kg
- b. 19.4 kg
- c. 14.29 kg
- d. 25.1 kg

199. On a psychrometric chart, cooling and humidification process appears as a line that is:

- a. diagonal upwards to the right
- b. diagonal downwards to the left
- c. diagonal upwards to the left
- d. diagonal downwards to the right

200. Shrimp harvested in a pond situated in Bohol are packed in insulated sheet metal containers and mixed with crushed ice while in transit to the shrimp processing plant in Manila. Initial temperature of the product is about 25°C and it is desired to cool the shrimp at approximately 0°C during shipment. Estimate the product load, in kJ/kg. For shrimp highest freezing temperature is -2.2°C, c_p above freezing is 3.38 kJ/kg-K, c_p below freezing is 1.79 kJ/kg-K and latent heat of fusion is 254 kJ/kg:

- a. 88.5
- b. 84.5
- c. 91.2
- d. 103.7